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Studierichting: Informatica

Oefeningenreeks 7E nr. 1.8

$$f(x, y) = 2x^2 + y^2 + 4x - 4y + 5$$

$$\frac{\partial f}{\partial x} = 4x - 4 = 0 \Leftrightarrow x = -1$$

$$\frac{\partial f}{\partial y} = 2y - 4 = 0 \Leftrightarrow y = 2$$

Te onderzoeken punt: $(-1, 2)$

$$H(x, y) = \det Df(x, y) = \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8$$

$$H(-1, 2) = 8 > 0$$

$$\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} = 4, 2 > 0 \Rightarrow \text{minimum in } (-1, 2)$$

References